

RO-TextCNN Based MUL-MOVE-Net for Camera Motion Classification

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Abstract—Video auto-editing is a new application of artificial intelligence technology in the media industry. Camera motion is one of the critical characteristics of videos and is vital for shot arrangement. Rules based camera motion classification algorithms generalize poorly from one dataset to another. Machine learning or deep learning based algorithms have better cross-task performance. However, in order to remove the motion information in the foreground area, researchers have to use semantic segmentation neural networks that are computationally expensive. Existing foreground segmentation algorithms are only effective for samples with clear foreground areas. The salient areas generated by the saliency segmentation algorithms are not the same as the foreground areas in many cases. This paper proposes a novel deep learning based camera motion classification framework MUL-MOVE-Net, which is composed of multiple instantaneous motion networks MOVE-Net. In MOVE-Net, a lightweight RO-TextCNN module is proposed to learn multi-scale templates in the 1D angle histogram of optical flow information. Without using semantic segmentation network, the algorithm is capable of foreground fault tolerance while ensuring efficiency. For the experiments, a dataset MOVE-SET is constructed with more than 100,000 pairs of instantaneous camera motion samples. On the testing set, our algorithm achieves an accuracy of 95.3% and a Macro-F1 value of 0.9385. In the shot-level motion classification task, the accuracy of MUL-MOVE-Net gets 4% higher than that of SGNet, and Macro-F1 0.3 higher. As a result, MUL-MOVE-Net could efficiently classify the camera motion in real time and is helpful for video auto-editing.

Index Terms—RO-TextCNN, MOVE-Net, Camera motion classification, Video auto-editing

I. INTRODUCTION

In recent years, the development of artificial intelligence has driven editing towards automation and intelligence. Video auto-editing is a new application of AI technology in the media industry. A machine needs to learn how to understand videos and how to select and sort these videos artistically. As one of the critical characteristics of videos, camera motion is an important reference of shots arrangement in video editing. Camera motion is represented by the global motion of the

video. The direction of the global motion and the camera motion is opposite. When the sample has less global motion, it is classified as static (S). Non-static samples are generally divided into eight categories, namely Up (U), Down (D), Left (L), Right (R), Zoom In (ZI), Zoom Out (ZO), Right Rotation (RR), Left Rotation (LR). Camera motion tasks include instantaneous (between two frames) motion classification and shot-level motion classification. The former is the basis of the latter.

There have been some researches on camera motion classification tasks. After extracting the motion information (motion vector [1] or optical flow [2]), some algorithms fit motion parameter models based on these motion information. Among the 4-parameter, 6-parameter and 8-parameter motion models, the most commonly used is the 6-parameter affine model [9] [10]. The six parameters in this model are directly related to global motion translation, scaling, and rotation. Some other algorithms directly count the optical flow vectors and set rules for classification rather than calculate the motion parameter models. In order to remove the optical flow vectors in the foreground area, some algorithms refer to the vision rules [3] or the motion information chaos rules. The latest algorithm in [8] does not remove foreground and noise but recognizes S/U/D/L/R by 2D optical flow histograms based rules and recognizes zoom and rotating by manual template matching. Over time, machine learning and deep learning have been widely used to classify camera motion. In recent years, optical flow 1D amplitude histograms and 1D angle histograms are often used as features by machine learning and deep learning algorithms [4] [6] [7]. Some algorithms such as [4] [5] also use semantic segmentation networks to segment background regions. In [5], to classify the camera motion of the entire shot, the Subject Guidance Network(SGNet) is used to guide the extraction of image features. Paper [5] also proposed a variance map to classify camera motion by the similarity of different video clips.

The motion parameter models based algorithms are suitable for accurate camera pose estimation, not qualitative camera motion classification tasks. Other traditional algorithms generalize poorly from one dataset to another. Algorithms based on machine learning or deep learning are highly transferable. However, in such algorithms, the background segmentation networks increases the complexity and computational complexity. The semantic segmentation algorithm used in [4] can only segment a few types of foreground objects. The salient areas obtained by the saliency segmentation algorithm in [5] is not equal to foreground areas in many cases. Our research aims to achieve tolerance to foreground and noise without semantic segmentation. We propose a lightweight deep network MOVE-Net to classify instantaneous motion. The RO-TextCNN module automatically learns multi-scale templates that can tolerate a small amount of foreground and outliers, achieving the best performance on our instantaneous motion dataset MOVE-SET. Compared with SGNet [5], the MOVE-Net based MUL-MOVE-Net framework also shows a significant performance improvement in the shot-level motion classification task of MovieShot.

The structure of this paper is as follows. In Chapter 2, we introduce MUL-MOVE-Net, a shot motion classification framework based on MOVE-Net. In Chapter 3, MOVE-Net and MUL-MOVE-Net are compared with other latest algorithms respectively on the instantaneous motion dataset MOVE-SET and the shot-level motion dataset MovieShot. Chapter 4 is the conclusion.

II. MUL-MOVE-NET

A. MUL-MOVE-Net

The proposed framework MUL-MOVE-Net for camera motion classification is shown in Fig1. The first step is to perform multi-point sampling. Divide a shot into n_{clips} clips, and take a pair of frame samples with an interval of d in each clip. MOVE-Net extracts amplitude information and angle information of optical flow for each pair of frames. In the consensus module, a single layer of biLSTM is used to process all features because the classification of shots needs the temporal information of different sampling points. Then the output of each timestep is averaged. Finally, the classification result is obtained through the FC layer. MOVE-Net will be introduced in detail below.

B. Extract the optical flow field

In MOVE-Net, the two input frames f_1, f_2 have the same size $3 \times h \times w$, where 3 denotes the RGB channel of the image. Liteflownet [2] is used to calculate the dense optical flow between these two frames. Other algorithms can also be used to calculate optical flow. The size of the extracted optical flow field is $3 \times h \times w$. Due to the continuity of the optical flow information at adjacent positions, the bilinear sampling method is used to downsample the optical flow field to another scaled optical flow field F of size $2 \times h \times w (l < h, l < w)$. The new optical flow field F consists of the optical flow F^x in the x direction and the optical flow F^y in the y direction.

This is to reduce the computational cost and remove redundant information without losing optical flow information.

Before the following calculations, we divide the optical flow field into nn regions. $n = 3$ in Fig1. Let $F_{i,j}(F_{i,j}^x, F_{i,j}^y)$ denote a pair of optical flow vectors at position (i, j) in F . Since the background is more likely to appear around the screen, in the calculation process below, we only take the optical flow information in the surrounding $m(m = 4n - 4)$ regions into consideration.

C. Optical flow histograms

In this step, the nonuniform 1D optical flow amplitude histogram and the 1D optical flow angle histogram of every region are extracted. The former can be used to judge whether a sample is static, while the latter is related to the specific motion types.

The nonuniform 1D amplitude histogram only counts the optical flow vectors in the surrounding m regions in the screen. First, calculate the optical flow amplitude matrix M corresponding to the optical flow field F by equation (1). The dimension of the constructed amplitude histogram is $n_m(n_m \leq 5)$. The length of the amplitude range of the first $n_m - 1$ bins of the histogram is 1 unit, and the amplitude range of the last bin is $(n_m - 1, +\infty)$. To construct this nonuniform amplitude histogram, we ignore the amplitude distribution at large amplitude to minimize the redundancy of features and prevent the model from overfitting. As shown in Fig2, the two amplitude histograms belong to a static sample and a non-static sample respectively. Finally, after the amplitude histogram is normalized, the final amplitude histogram H_m is obtained.

$$M_{i,j} = \sqrt{F_{i,j}^x{}^2 + F_{i,j}^y{}^2} \quad (1)$$

As for the angle histogram, according to the commonly used angle division rule in Fig3, the normalized 1D angle histogram $H_a^u \in R^{1 \times 8} (u \in 1, 2, \dots, m)$ is calculated for the m regions around the screen. Fig4 are the examples of standard angle histograms for eight types of camera motion. The angle histogram sequence $H_a \in R^{m \times 8}$ is obtained by concatenating H_a^u .

D. RO-TextCNN

Instead of output H_a as the final angle feature, the template (convolution kernel) matching feature $\widehat{H}_a$ is further extracted for H_a . In text classification tasks, TextCNN [11] uses convolution kernels with different window sizes to extract features of different scales in the text. The angle histogram sequence H_a is end-to-end, so we propose a rotating TextCNN (RO-TextCNN) module, which uses different sizes of convolution kernels (templates) to extract diverse matching information from end-to-end sequences.

Three convolution kernels with different window sizes are designed as Fig1. The first type of convolution kernel has the window size of n , the step size of $n - 1$, and a number of convolution kernels of n_1 . The convolution kernel extracts the matching features of the four sides in H_a , which is

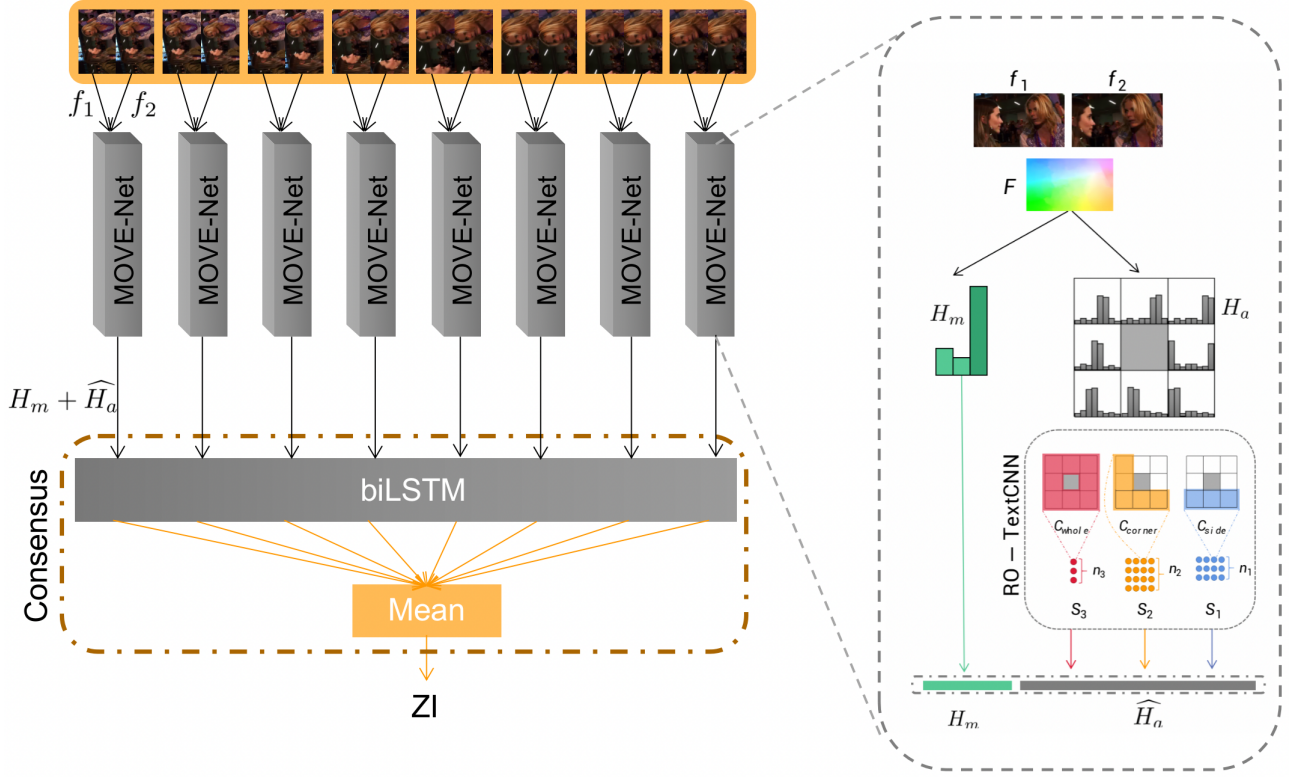


Fig. 1. The structure of MUL-MOVE-Net.

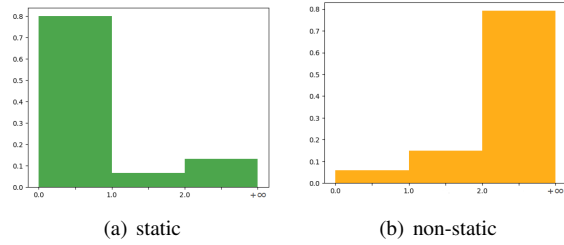


Fig. 2. Nonuniform amplitude histograms of static and non-static samples.

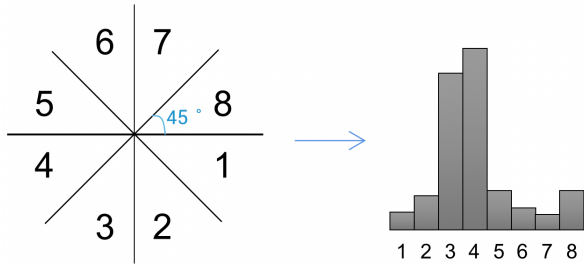


Fig. 3. Angle division rule.

called C_{side} convolution. The convolution result $S_1 \in R^{n_1 \times 4}$ is obtained. The second type of convolution kernel has a window size of $2n - 1$, a step size of $n - 1$, and a number

of convolution kernels of n_2 . This convolution extracts the matching features of the four corners in H_a , which is called C_{corner} convolution. The convolution result $S_2 \in R^{n_2 \times 4}$ is obtained. The third convolution kernel has a window size of m . That is, convolve the entire H_a . The number of convolution kernels is n_3 . This convolution is called C_{whole} convolution, and the convolution result $S_3 \in R^{n_3 \times 1}$ is obtained. The three results are flattened and concatenated, and the template matching feature $\hat{H}_a \in R^{4n_1 + 4n_2 + n_3}$ is obtained.

This module satisfies the human eye's perception of camera motion. That is, pay attention to the macro motion patterns and ignore the small amount of outlier motion details. What is matched in the RO-TextCNN module is a group of regions on the sides or corners rather than a single region. Therefore, when foreground objects or noises are in a few regions, the proposed algorithm has an excellent fault-tolerant ability, and the trained template can also better resist interference.

III. EXPERIMENT AND ANALYSIS

To verify the performance of our proposed camera motion classification framework MUL-MOVE-Net, the experiments in this chapter are designed. The experiment is carried out in a 12G GTX 1080Ti GPU accelerated environment, and the model is built by PyTorch.

A. Experiment I for MOVE-Net

Experiment I verifies the performance of the instantaneous motion network MOVE-Net. In this experiment, the output

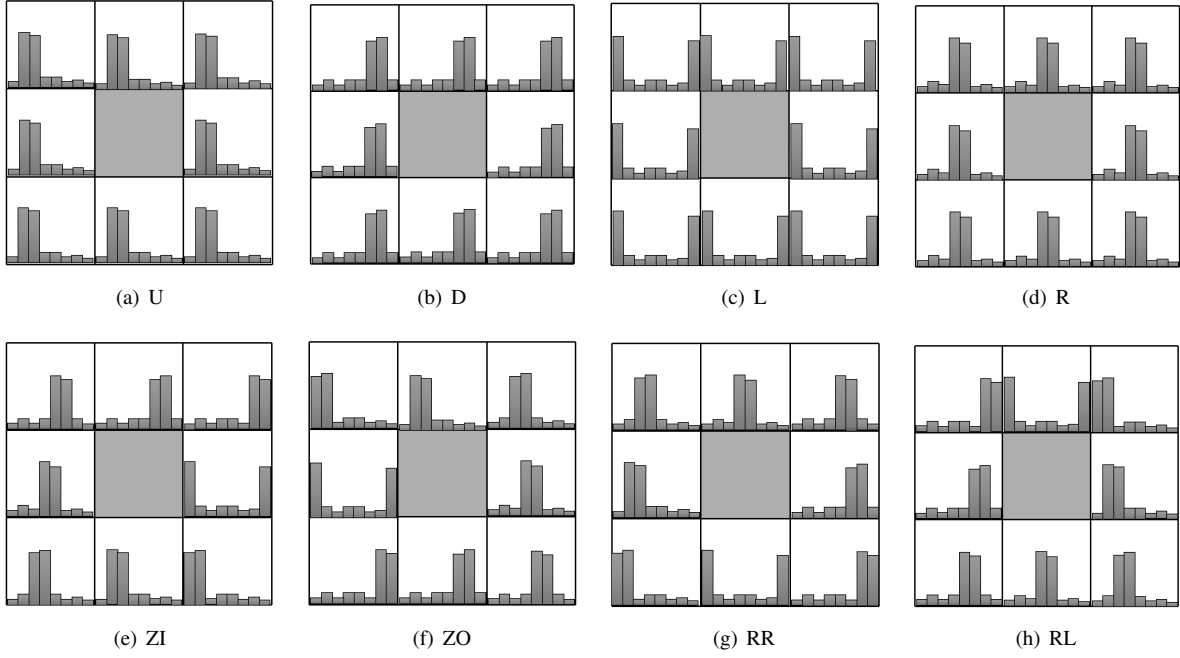


Fig. 4. Standard angle histograms of 8 types of camera motion.

feature $(H_m + \widehat{H}_a)$ of MOVE-Net is input to an FC layer for instantaneous camera motion classification and compared with the latest instantaneous motion classification algorithms. The experiment is carried out on a new instantaneous camera motion dataset MOVE-SET. The material sources of this dataset include various video types such as ceremonies, news, variety shows, concerts, film and TV dramas, sports programs, cartoons, and live videos. We use the shot boundary detection algorithm of [12] to cut the video shot. The annotators mark the smooth motion segments of some shots. The instantaneous motion category at each moment in the smooth motion segment is the same, making it more convenient for us to construct instantaneous motion sample pairs at intervals of d frames. The dataset is divided into a training set and a testing set, and the ratio of the number of samples is about 3:1. The total number of videos and shots are shown in Table I, and the number of sample pairs of each type of camera motion is shown in Table II.

TABLE I
TOTAL NUMBER OF VIDEOS/SHOTS IN MOVE-SET

Dataset	videos	shots
Training set	300	1213
Testing set	148	490

In the experiment, the latest algorithms in paper [8] and [4] are compared with MOVE-Net. In the three algorithms, the frame interval d is set to be 6. In MOVE-Net, the side length l of the scaled optical flow field F is set to be 32. As in paper [4], F is divided into 33 regions. When calculating the amplitude histogram, the value of n is 3. During training,

TABLE II
THE NUMBER OF SAMPLE PAIRS FOR EACH MOTION TYPE IN MOVE-SET

Dataset	U	D	L	R	S
Training set	5001	4875	5805	5936	24648
Testing set	1594	1675	2515	2612	11577
Dataset	ZI	ZO	RR	RL	
Training set	11000	6111	3290	4357	
Testing set	3357	1722	1415	1746	

the learning rate is set to 0.004, and the Adam optimizer is used. Set momentum parameters $\beta_1 = 0.9, \beta_2 = 0.999$. The *batchsize* is 32. Each model is trained on the training set for eight epochs, and then the accuracy ACC and macro-F1 values of different models are compared on the testing set.

Table III is the result of experiment I. As shown in the table, the accuracy and Macro-F1 of [8] are about 16% and 0.21 lower than our algorithm. Since our algorithm is based on deep learning, its performance is better than traditional algorithms that are not flexible. Compared with the algorithm in [4], MOVE-Net integrates amplitude information and angle information for end-to-end training. RO-TextCNN is also used for multi-scale template matching to resist foreground and noise, achieving a higher accuracy of 95.3% and a macro-F1 value of 0.9385.

B. Experiment II for MUL-MOVE-Net

Experiment II verifies the performance of the shot-level motion framework MUL-MOVE-Net. The experiment is carried out on the shot-level motion dataset MovieShot [5]. The dataset contains about 33,000 shots. The camera motion is divided into static (S), general motion (M), push (PUSH), and

TABLE III
THE RESULT OF EXPERIMENT I

Algorithm	ACC	Macro-F1
Prasertsakul, P., et al. [8]	79.45%	0.7212
Derue, F. X., et al. [4]	83.80%	0.9044
MOVE-Net	95.30%	0.9385

pull (PULL). Among them, there are much more S and M shots than PUSH and PULL shots.

The algorithm compared with MUL-MOVE-Net is the latest SGNet [5]. In addition, we also replaced the consensus module in MUL-MOVE-Net with the same averaging method as SGNet, as an intermediate experiment (MUL-MOVE-Net(without biLSTM)). We take $n_{clips} = 8$. Other parameters remain consistent with the experiment I. Table IV shows the accuracy ACC and macro-F1 values of each framework on MovieShot.

TABLE IV
THE RESULT OF EXPERIMENT II

Framework	ACC	Macro-F1
SGNet(Var-img) [5]	78.91%	0.3823
MUL-MOVE-Net(without biLSTM)	82.05%	0.6873
MUL-MOVE-Net(with biLSTM)	82.85%	0.6910

As shown in Table III, the Macro-F1 value of SGNet is 0.3823, which indicates that SGNet's ability to classify various categories is not balanced. Compared with the variance map in SGNet, MOVE-Net in MUL-MOVE-Net can integrate amplitude and angle information to avoid this imbalance, so the Macro-F1 increases significantly to 0.6873. When biLSTM is added to the consensus module, MUL-MOVE-Net(with biLSTM) achieves 82.85% accuracy and 0.691 Macro-F1 due to the processing of temporal information.

In our experimental environment, the proposed algorithm can process 20 pairs of instantaneous samples per second, while algorithms [5] and [4] can only process about 14 pairs. A deep learning framework based on semantic segmentation not only has a certain probability to segment the wrong foreground, but also has a higher computational cost. This paper is dedicated to designing a smaller network to reduce the interference of foreground and noise. Experiments show that our algorithm achieves the best performance.

IV. CONCLUSION

As one of the key characteristics of videos, camera motion is an important reference of shots arrangement in video editing. The existing camera motion classification algorithms have some problems, including poor transferability, poor anti-interference ability to the foreground, high computational cost, and so on. To this end, this paper proposes a new camera motion classification framework MUL-MOVE-Net. The core of the framework is the lightweight instantaneous motion network MOVE-Net. In MOVE-Net, the nonuniform 1D amplitude histogram optimizes the judgment accuracy

of static categories, and the multi-scale automatic template matching module RO-TextCNN reduces the interference of foreground and noise on classification. MOVE-Net achieves 95.3% accuracy and 0.9385 Macro-F1 value on the instantaneous camera motion dataset MOVE-SET, which contains about 100,000 pairs of instantaneous motion samples. The performance of MOVE-Net is better than other instantaneous motion classification algorithms in recent years. MUL-MOVE-Net, a shots-level motion classification framework based on MOVE-Net, achieves an accuracy of 82.85% and a Macro-F1 value of 0.691 on the MovieShot dataset. The performance is better than SGNet.

Our algorithm also has some shortcomings to be solved. In MOVE-Net, The fault tolerance of RO-TextCNN is limited. When the foreground area extends to the edge of the screen, errors in the classification may occur. Lightweight networks need to be developed to resist interference from foreground and noise robustly. This is what we will explore in the future.

V. ACKNOWLEDGMENT

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